

Student Answer Script View



MIT Bangalore MID-TERM 2025-2026

Answer Sheet

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Comments submitted by student.

Q.No : 1)		Score : 1.00 / 1.00
Which of the following best describes the difference in error signals between supervised learning (SL) and reinforcement learning (RL)?		
Supervised learning relies on sparse, delayed error signals, while RL uses immediate and detailed labels.	Both rely on the same gradient-based error propagation from labelled targets.	Supervised learning uses explicit error from labels, while RL uses implicit feedback in the form of rewards or penalties.
		RL agents can compute exact loss values like cross-entropy after every action.

Q.No : 2)

Score : 0.00 / 1.00

Suppose two models are compared: Model A minimizes MSE, Model B minimizes MAE. On a dataset with heavy outliers, which of the following is true?

Model A's parameters will shift more toward fitting the outliers.

Model B's parameters will overfit the outliers more strongly.

Both models will ignore the outliers equally.

Model A will approximate the median, while Model B approximates the mean.

Q.No : 3)

Score : 1.00 / 1.00

In terms of achieving generalization, which statement best explains the bias-variance trade-off?

Reducing bias always reduces variance.

Increasing variance always improves generalization.

A balance between bias and variance minimizes expected test error.

Bias and variance are unrelated to test performance.

Q.No : 4)

Score : 1.00 / 1.00

When comparing discrimination and regression approaches to classification, which is true?

Regression always produces linear decision boundaries, while discriminant analysis can produce nonlinear ones.

Discriminant functions explicitly model class boundaries, while regression functions approximate target values and thresholds are imposed afterward.

Regression requires probability density assumptions, while discriminant functions do not.

Both are equivalent whenever the classes are balanced.

Q.No : 5)

Score : 0.00 / 1.00

Which statement best explains the relationship between gradient descent and the cost function?

Gradient descent updates parameters in the direction of the gradient to maximize the cost function.

Gradient descent uses the negative gradient of the cost function to iteratively reduce its value.

Gradient descent moves orthogonally to the gradient to avoid saddle points.

Gradient descent guarantees reaching the global minimum of any cost function.

Q.No : 6)

Score : 1.00 / 1.00

Suppose logistic regression is used with

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

If

$$h_{\theta}(x) = 0.9$$

then the log-odds are approximately:

0 1 2.2 9

Q.No : 7)

Score : 0.00 / 1.00

The residual error in linear regression is defined as:

$$e^{(i)} = y^{(i)} - \hat{y}^{(i)}$$

This quantity is minimized across all samples by:

Maximizing XX^T .

Adjusting weights randomly until convergence.

Replacing regression with classification.

Minimizing the residual sum-of-squares criterion.

Q.No : 8)

Score : 1.00 / 1.00

For discriminant function

$$g(x) = -x_1 + 2x_2 - 5,$$

determine the side of the boundary for point

$$(x_1 = 4, x_2 = 3).$$

 $g(x) = 1$, Class 1 $g(x) = -3$, Class 2 $g(x) = 0$, On boundary $g(x) = 7$, Class 1 side

Q.No : 9)

Score : 1.00 / 1.00

The Lagrangian for the hard-margin SVM introduces multipliers.

$$\lambda_i \geq 0.$$

Their analytical role is:

To weight the contribution of misclassified points.

To ensure $w^T w$ remains convex.

To regularize against overfitting.

To enforce linear separability constraints during optimization.

Q.No : 10)

Score : 1.00 / 1.00

Consider the hyperplane

$$3x_1 - 4x_2 + 12 = 0$$

The distance of the origin from this hyperplane is:

☒ 12/5☐ 3/5☐ 4/5☐ 7/5

Q.No : 11 A)**Score : 1.50 / 4.00**

Consider two datasets: One that is perfectly linearly separable, and another where classes overlap due to noise.

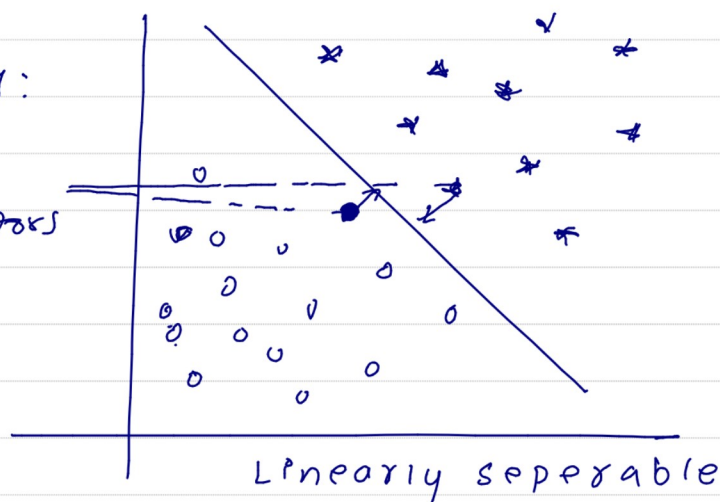
- a. Analyze which classifier (the maximal margin classifier or the soft margin classifier) would be more suitable for each dataset.
- b. Explain the geometric reasoning behind the choice (margin width, feasibility, robustness).
- c. Discuss the role of slack variables and the parameter C in the case of overlapping classes.

Illustrate with clear diagrams showing the separating hyperplane, margin boundaries, and support vectors.

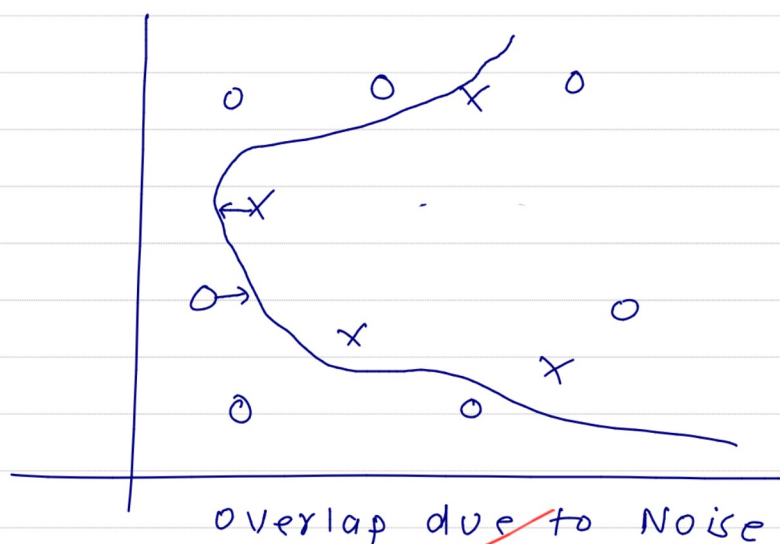
Solution:-

dataset 1:

support
vectors



Dataset 2 :



- a) For the first dataset (D_1), maximal margin classifier would be the most suitable for the dataset (D_1)

For second dataset (D_2) soft margin classifier would be suitable for dataset (D_2)

- b) The geometric reasoning is because maximal margin width supports vectors can be easily used in this situation to perform classifications, on the otherhand

for dataset (D2) the robustness
allows the classification and overcome
the noise in the case of soft margin

Q.No : 11 B)		Score : 0.00 / 3.00
An underwater monitoring system analyzes 800 hydrophone clips to detect illegal blast fishing. • 12.5% of the clips contain a blast. • The system predicts "Blast" for 18% of the clips. • Of these predicted positives, 72 are correct detections. Using the given information, construct the confusion matrix. Apply the formulas for Precision, Sensitivity (True Positive Rate), and Specificity (True Negative Rate) to this scenario. Compute the F1 score using the obtained values.		

Given:- $N = 800$ hydrophone clips

$$N(\text{contain blast}) = \frac{12.5}{100} \times 800 = 100 \text{ clips}$$

$$N(\text{predicted blast}) = \frac{18}{100} \times 800 = 144 \text{ clips}$$

$$N(\text{correction detection}) = 72$$

confusion matrix.

	Actual classifier +ve	Actual classifier -ve
Classifier +ve	TRUE PREDICTION = 100 clips	FALSE NEGATIVE = 44 clips
Classifier -ve	FALSE PREDICTION = 144 clips	TRUTH NEGATIVE = 72 clips

$$\text{precision} = \frac{TP + TN}{TP + FP + FN + TN} = \frac{100 + 72}{366}$$

$$\text{precision} = \frac{172}{366} = 0.477$$

$$\text{sensitivity} = \frac{TP}{TP + FN} =$$



Q.No : 11 C)

Score : 0.50 / 3.00

Analyze how Ant Colony Optimization (ACO) and Particle Swarm Optimization (PSO) differ in their sources of biological inspiration and the way collective behaviour is modelled. Highlight at least one similarity and one difference in their mechanisms.

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Solution:-

Ant colony optimization	Particle swarm optimization
i) This is a <u>metaheuristic algorithm</u> , that uses graphs to find out the optimal path to the solution just like the ant colony	i) This is a algorithm that <u>uses trees</u> to find the optimal path to the solution of a given problem as a particle swarms
ii) similarity is both are used to find the optimal path in a metaheuristic way	
iii) uses graphs as its datastructure	iii) uses trees as its datastructure
iv) complexity $O(V+E)$	complexity $O(V+E)$

Q.No : 12 A)

Score : 5.00 / 5.00

You are given the following training dataset for predicting house categories:

ID	x	y	Category
A	7	8	High
B	6	6	Medium
C	5	9	Medium
D	9	8	High
E	3	4	Low

Two test points are to be classified:

1. $T_1 = (5,7)$
2. $T_2 = (7,6)$

Apply the k-NN classification algorithm to compute the predicted category for the two test points, T_1 and T_2 with Manhattan distance and $k=3$.

ID	x	y	category
A	7	8	High
B	6	6	medium
C	5	9	medium
D	9	8	High
E	3	4	Low
T ₁	5	7	unknown
T ₂	7	6	unknown

Manhattan distance

$$d(x, y) = \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{1/p}$$

When $p=1$ it becomes
Manhattan distance

$$d(x, y) = \sum_{i=1}^n |x_i - y_i|$$

For T₁ point T₁ (5, 7) let us calculate
the distances x₁ y₁

For pt A (7, 8) and T₁ (5, 7)

$$\begin{aligned} d(A, T_1) &= |7-5| + |8-7| \\ &= 2 + 1 \\ &= 3 \end{aligned}$$

$$\begin{aligned} d(B, T_1) &= |6-5| + |6-7| \\ &= 1 + 1 \\ &= 2 \end{aligned}$$

$$\begin{aligned} d(C, T_1) &= |5-5| + |7-9| \\ &= 0 + 2 \\ &= 2 \end{aligned}$$

$$\begin{aligned} d(D, T_1) &= |5-9| + |7-8| \\ &= 4 + 1 \\ &= 5 \end{aligned}$$

$$\begin{aligned} d(E, T_1) &= |5-3| + |7-4| \\ &= 2 + 3 \\ &= 5 \end{aligned}$$

arrange the distances in ascending order.

ID	Manhattan distances	categories
B	2	Medium
C	2	Medium
A	3	High
D	5	High
E	5	Low

for $K=3$

∴ For point T_1 its classified in Medium

For Point $T_2 = (7, 6)$ manhattan distance

$$\text{For } d(x, y) = \left(\sum_{i=1}^n (|x_i - y_i|^p) \right)^{1/p} \text{ when } p=1$$

$$\begin{aligned} d(A, T_2) &= |7-7| + |6-8| \\ &= 0 + 2 \\ &= 2 \end{aligned}$$

$$\begin{aligned} d(B, T_2) &= |7-6| + |6-6| \\ &= 1 + 0 \\ &= 1 \end{aligned}$$

$$\begin{aligned} d(C, T_2) &= |7-5| + |6-9| \\ &= 2 + 3 \\ &= 5 \end{aligned}$$

$$\begin{aligned} d(D, T_2) &= |7-9| + |6-8| \\ &= 2 + 2 = 4 \end{aligned}$$

$$\begin{aligned}d(E, T_2) &= |7 - 3| + |6 - 4| \\&= 4 + 2 \\&= 6 \checkmark\end{aligned}$$

ID	Manhattan distance	categories
B	1	Medium
A	2	High
D	3	High
C	5	Medium
E	6	Low

For $k=3$

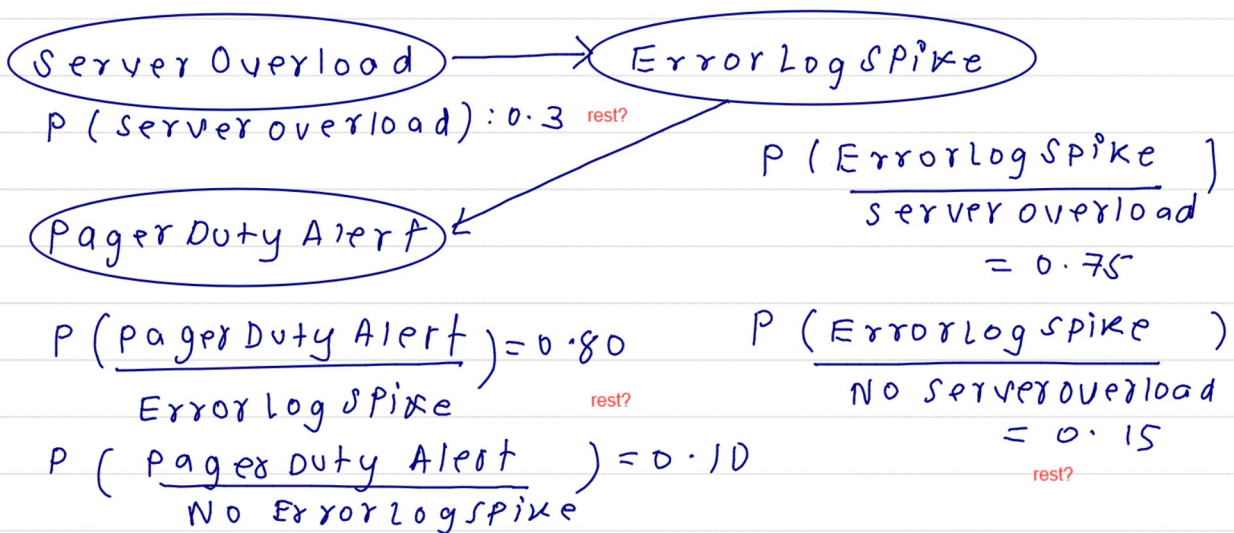
For point T_2 is classified as High.

Q.No : 12 B)**Score : 0.50 / 3.00**

Apply the law of total probability in a Bayesian Belief Network with nodes ServerOverload, ErrorLogSpike, and PagerDutyAlert. ServerOverload influences whether an ErrorLogSpike occurs. ErrorLogSpike in turn influences whether a PagerDutyAlert is triggered. The network parameters are:

- The probability of ServerOverload is 0.3.
- The probability of ErrorLogSpike given ServerOverload is 0.75.
- The probability of ErrorLogSpike given no ServerOverload is 0.15.
- The probability of PagerDutyAlert given ErrorLogSpike is 0.80.
- The probability of PagerDutyAlert given no ErrorLogSpike is 0.10.

Draw the Bayesian Belief Network diagram clearly showing the nodes and directed edges. Construct the Conditional Probability Tables (CPTs) for ServerOverload, ErrorLogSpike and PagerDutyAlert. Compute the probability of PagerDutyAlert given ServerOverload.



$$P\left(\frac{\text{Pager Duty Alert}}{\text{Server Overload}}\right) = ? \quad TP + PN$$

$$P\left(\frac{\text{Pager Duty Alert}}{\text{Server Overload}}\right) = \frac{P\left(\frac{\text{Server Overload}}{\text{Pager Duty Alert}}\right) P(\text{Server Overload})}{P\left(\frac{\text{Server Overload}}{\text{Pager Duty Alert}}\right) \cdot P(\text{Server Overload}) + P\left(\frac{\text{No Server Overload}}{\text{Pager Duty Alert}}\right) \cdot P(\text{No Server Overload})}$$

$$P\left(\frac{\text{Server Overload}}{\text{Pager Duty Alert}}\right) \cdot P(\text{Server Overload}) +$$

$$P\left(\frac{\text{No Server Overload}}{\text{Pager Duty Alert}}\right) \cdot P(\text{No Server Overload})$$



Q.No : 12 C)

Score : 0.50 / 2.00

Evaluate the importance of the learning rate in the Perceptron update rule. What could happen if it is set too high or too low?

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Solution:- The perceptron update rule has immense importance:

(i) Whenever we use perceptron learning rate rule, our regression model data point weights get updated and are corrected using the perceptron update learning algorithm.

ii) When we set this too low for example as (epoch=1,2), there is a high chance that training of the dataset, will be less hence having lesser accuracy which may lead to underfitting

iii) If we set it too high, as epoch = (100, 20,000) the model will experience overfitting as there is no will be no scope to test fresh dataset excluding the training data.



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